Digital Signal Processing

Part I – Digital Signals: processing in computers and doing math

Topic I.1 – Analog, Digital and Discrete-Time Signals

Prof. Aldebaro Klautau, Ph.D.
Federal University of Pará (UFPA)

www.lasse.ufpa.br & www.ppgee.ufpa.br

Definition of signal and some examples

General definition: anything that carries information

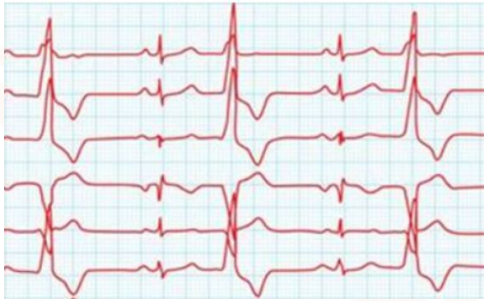
Image

Monochromatic images are 2D signals, information are the pixel values, there is no time evolution, example: 3D matrix (tensor) 480x640x3 (RGB color) **X**

Video

Sequence of images over time, example: 4D tensor 500x480x640x3 representing 500 frames **X**[n]

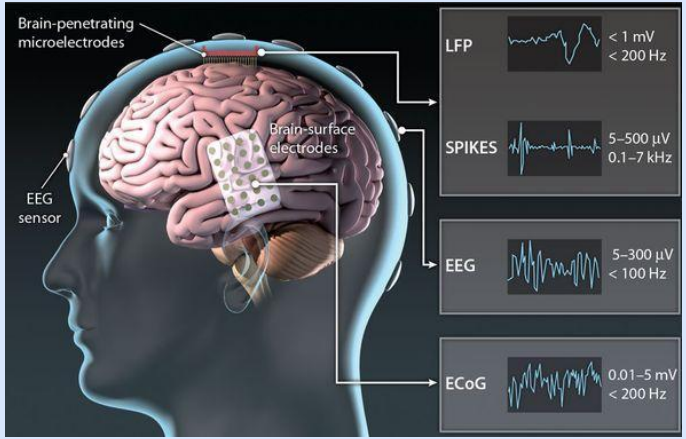
Electrocardiogram (ECG)



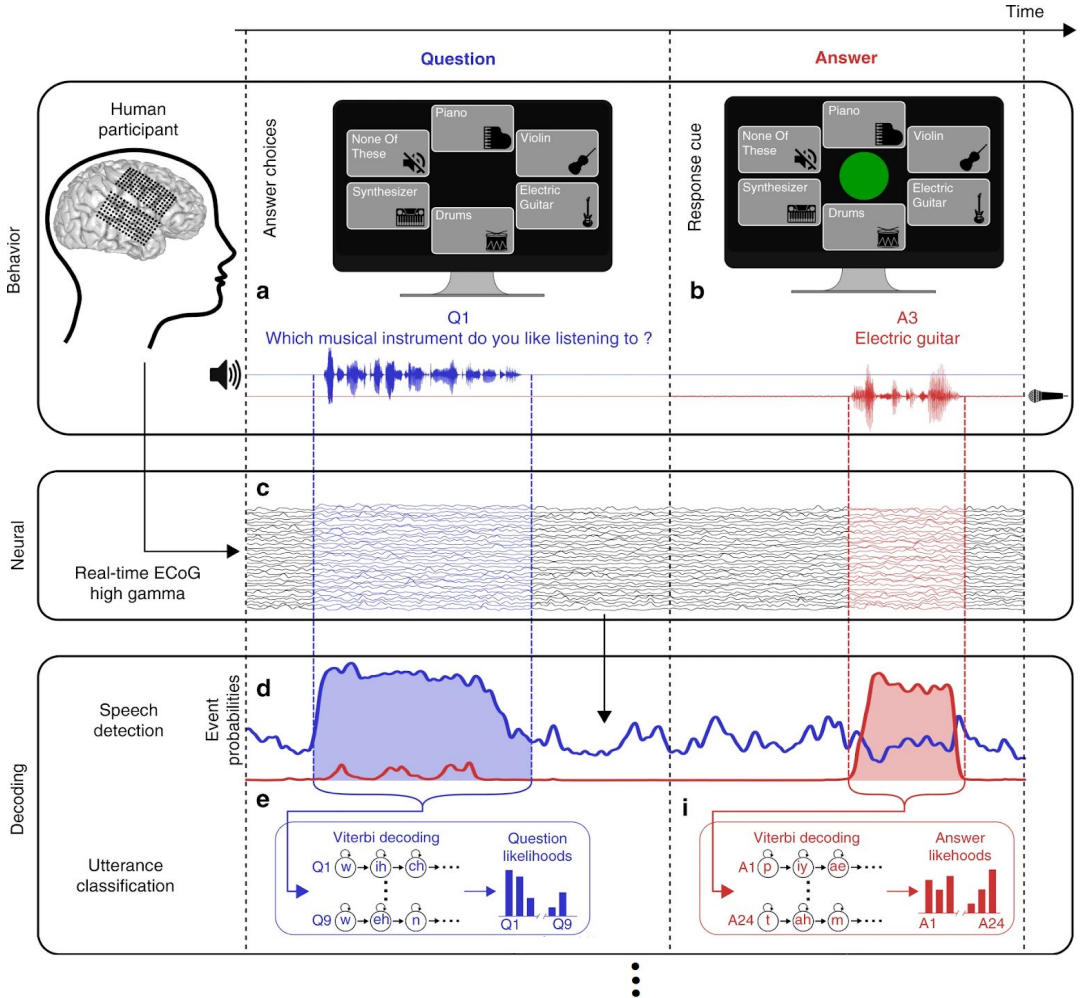
Multichannel ECG: Multivariate signal, information are amplitudes, example: 6 channels, with amplitudes in vector of dimension 6 **x**(t)

Example of modern digital signal processing: decoding brain signals

Electrocorticography (ECoG) vs electroencephalography (EEG)

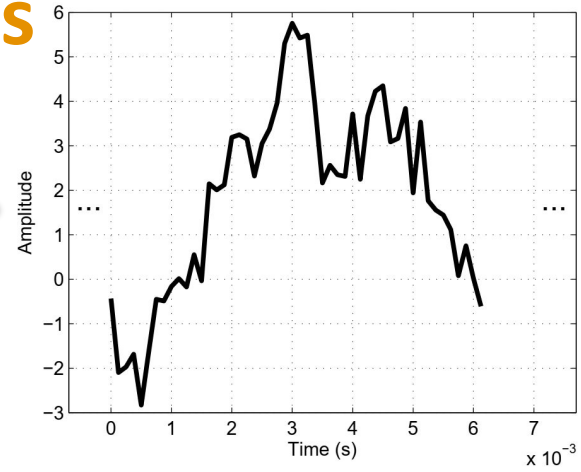


[Moses, 2019] https://www.nature.com/articles/s41467-019-10994-4



Real-valued 1D time-domain signals

For concreteness:
Single dimension
Information is real number



Time: continuous or discrete?

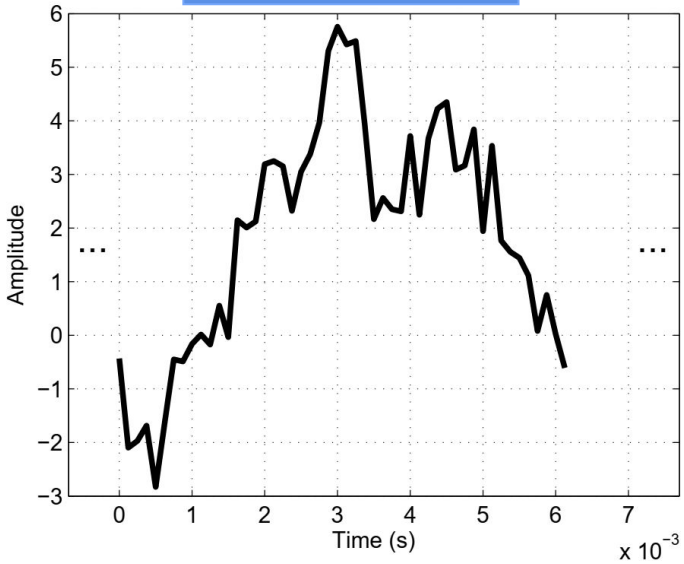
Types of signals

- If time evolution is represented by a real-valued variable $t \in \mathrm{R}$, the function $x(t)$ is called a *continuous-time signal*.
- If the progress over time is represented by an integer index $n \in \mathrm{Z}$, the sequence $x[n]$ is called a *discrete-time signal*.

Analog versus digital signals

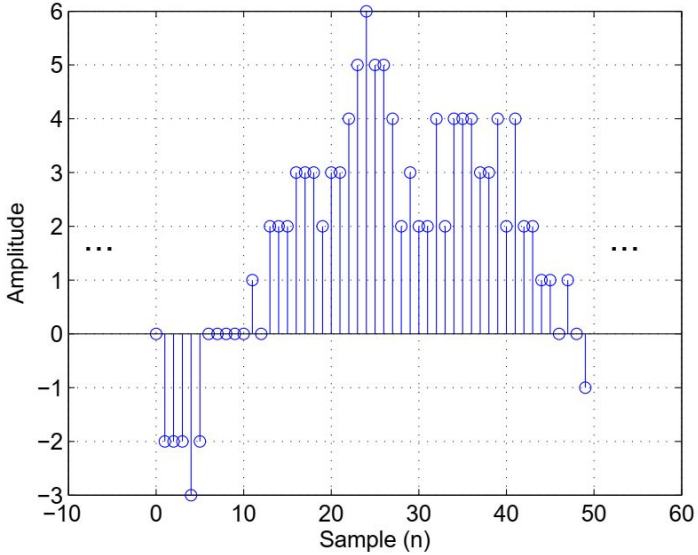
Digital signal: discrete in both time and amplitude
Analog signal: continuous in both (no restrictions)

$x(t)$ analog



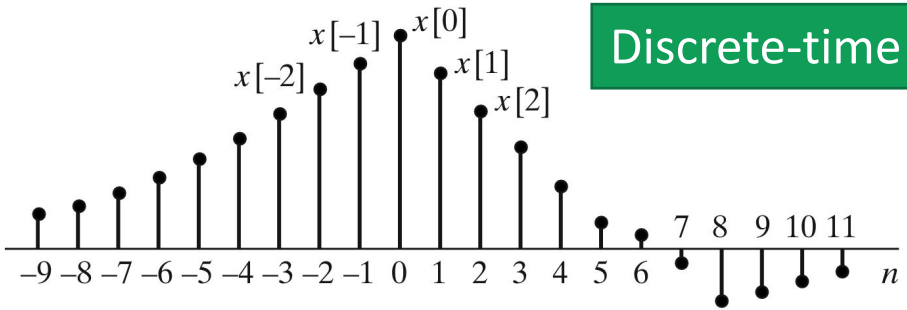
Amplitude quantized to 10 values in {-3, -4, ..., 5, 6} that could be represented with 4 bits

$x_q[n]$ digital



Four signal categories and notation

Table 1.1	Continuous-time, $t \in \mathrm{R}$	Discrete-time, $n \in \mathrm{Z}$
Quantized	$x_q(t)$	$x_q[n]$ (**digital**)
Not quantized	$x(t)$ (**analog**)	$x[n]$



Discrete-time signal

When doing math, we will be using **discrete-time** signals instead of digital signals

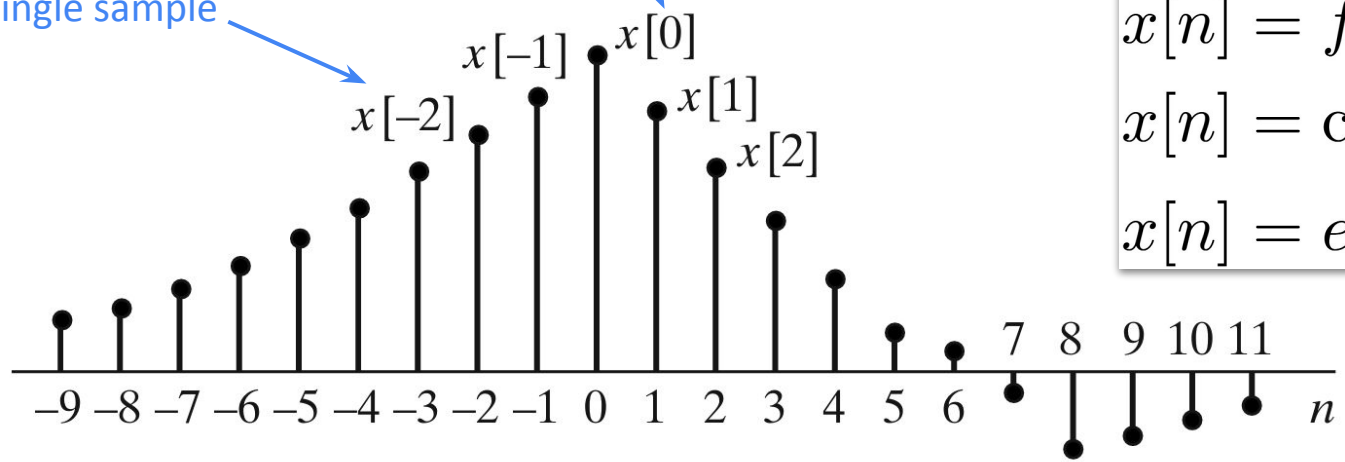
- The reason is that "quantization" is a non-linear operation that complicates incorporating, e.g., Fourier transforms and other operations

Beware: ambiguous notation!

The same notation $x[n]$ is used for both the whole sequence and for a specific sample at time *n*

Single sample

Single sample



Whole sequence:

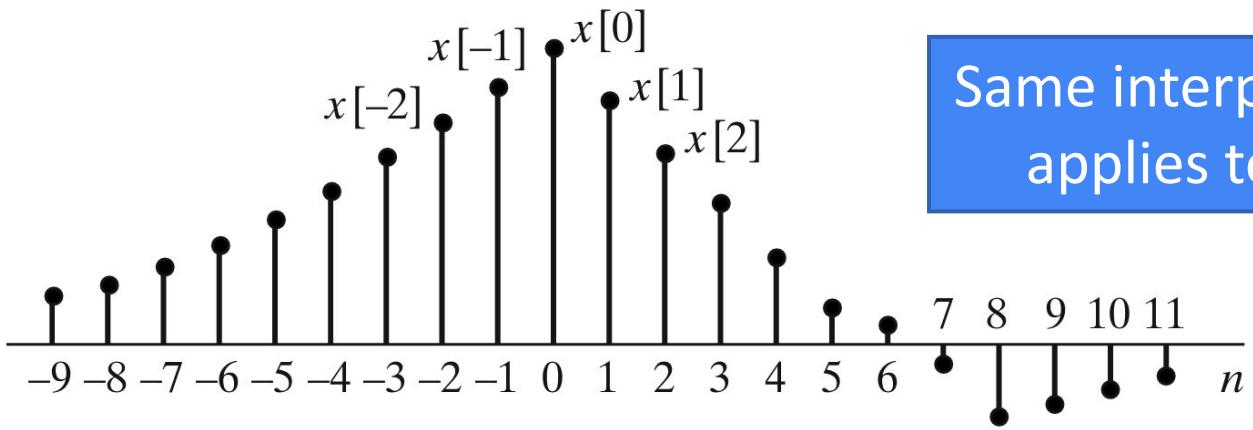
$$x[n] = f(n),$$
$$x[n] = \cos(3n),$$
$$x[n] = e^{j\Omega n}, \text{ etc.}$$

Notation: $x[n_0]$ (sample at time n_0) or $x[n]$ (sample or sequence?)

Notation (Our interpretation to its ambiguity)

We use n_0 or t_0 to denote a specific time instant

$x[n_0]$ is a sample at the given time n_0, but we do not assume we can identify this time instant. Example: when $x[8] = -2$, we interpret $x[8]$ as the number -2. We do not know that -2 is associated with n_0=8. Notation $x[n_0]$ there is no time information

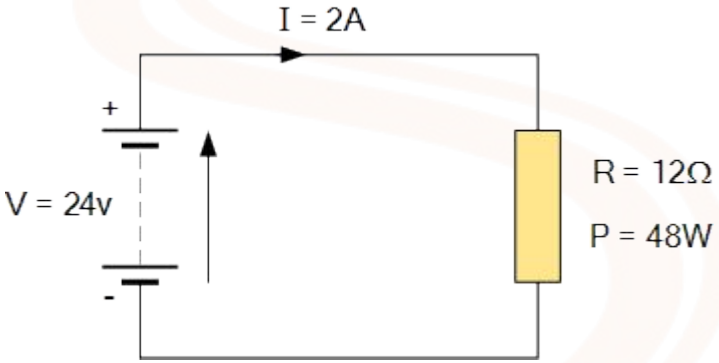


Same interpretation applies to $x(t_0)$

Convention: assume signal *x(t)* was obtained over a resistor with resistance = 1 Ω

Advantage: we can get additional insight when interpreting results and expressions using units such as Watts

Remind simple circuit in which power is $P = V^2/R = I^2 R$ Watts.
If R = 1 W, then: $P = V^2 = I^2$



In DSP, we often assume R = 1 Ω, such that the instantaneous power is

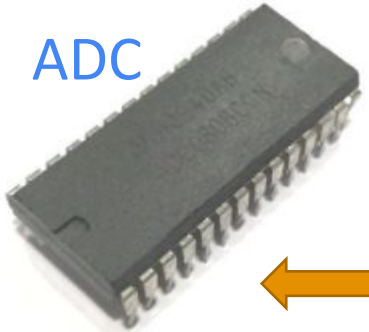
$$p(t) = x^2(t)$$

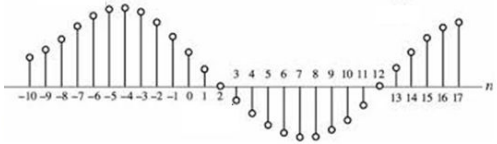
and we do not worry if $x(t)$ is in volt or ampere

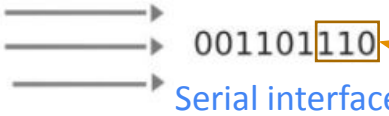
Obs: If the resistance is not 1 Ω, one can use "impedance scaling"

Analog-to-digital chip (ADC)
Digital-to-analog chip (DAC)

ADC





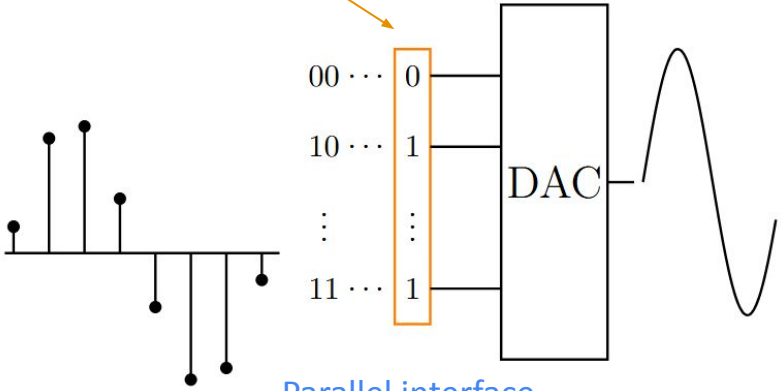


Example: ADC chip with F_s=100 Msps and b=8 bits/sample generates a signal with rate $R = F_s\, b$ = 800 Mbps

Single sample

Main parameters: sampling frequency F_s and number b of bits

Unit of sampling frequency F_s is Hertz (Hz) or, equivalently, samples per second (sps)



Manipulating the independent variable (*t* or *n*)

Manipulating signals in practice with softwares (Audacity, etc) ✓

Manipulating signals algebraically to do "theoretical" analysis ?

Original *x*(*t*) □ manipulation □ New signal y(t)

Very different from manipulating the dependent variable (amplitude)

For example, $y(t) = 4x(t)$ scales the amplitude (y-axis) by 4, while $y(t) = x(4t)$ contracts time (x-axis) by 4

One future example of signal manipulation:

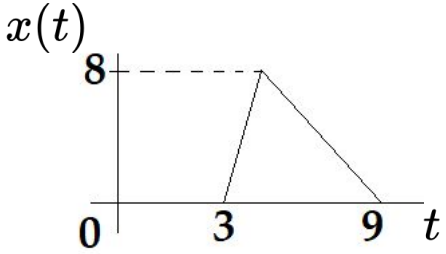
$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Useful to learn basic operations:

- Flip over y-axis: $y(t) = x(-t)$
- Time-shift: $y(t) = x(t-1)$
- Contraction: $y(t) = x(3t)$
- Dilation: $y(t) = x(t/2)$

Distinguish manipulations of independent and dependent variables

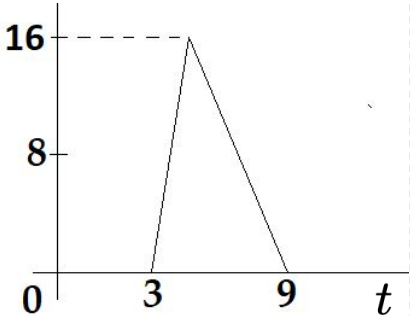
Original $x(t)$



Manipulating the dependent variable

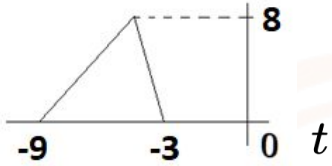
$y(t) = 2x(t)$

Modifies the y-axis



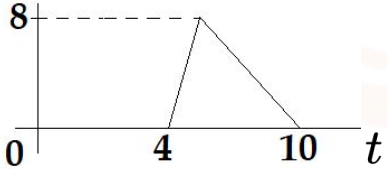
Manipulating the independent variable

$y(t) = x(-t)$



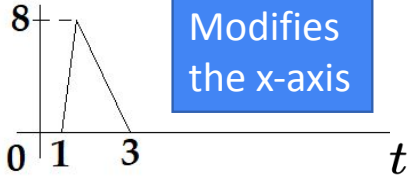
Flip over y

$y(t) = x(t-1)$



Time-shift

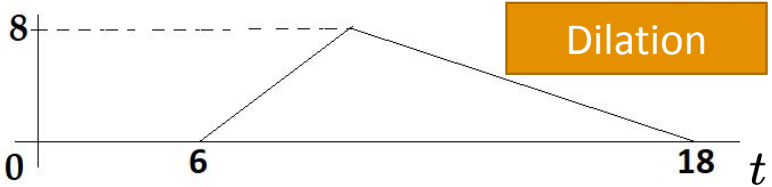
$y(t) = x(3t)$



Modifies the x-axis

Contraction

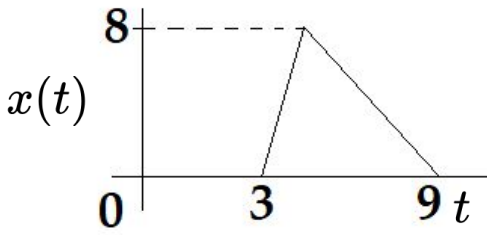
$y(t) = x(t/2)$



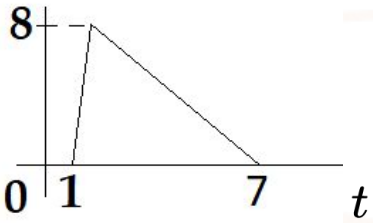
Dilation

Rule of thumb 1: where-to-shift

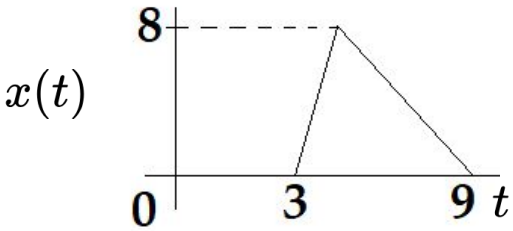
- assuming that $t_0 > 0$, $x(t - t_0)$ is a delayed version (right-shifted) of $x(t)$ and $x(t + t_0)$ is an anticipated version (left-shifted) of $x(t)$;



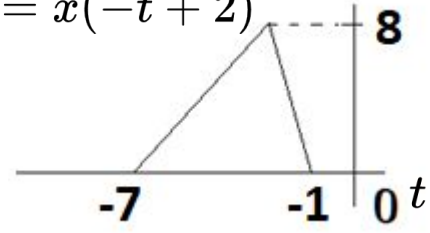
$y(t) = x(t + 2)$



- still assuming that $t_0 > 0$, $x(-t + t_0)$ is obtained by flipping $x(t)$ with respect to the y-axis and delaying $x(-t)$ by t_0, while $x(-t - t_0)$ is also obtained by flipping $x(t)$ but advancing $x(-t)$ by t_0.



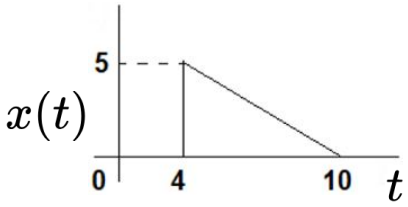
$y(t) = x(-t + 2)$

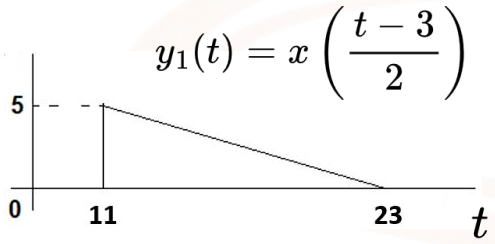


Do not:
$y(t) = x(-(t - 2))$
to obtain
$x(t - 2)$
and then flip

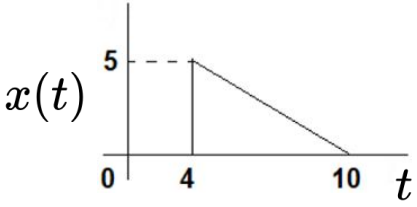
Rule of thumb 2: Simultaneous scale & shift

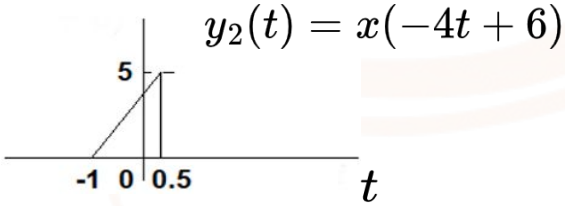
- for $x((t+\gamma)/\xi)$, first expand or contract $x(t)$ by a factor of ξ and then shift the intermediate result by γ;





- for $y(t) = x(\alpha t + \beta)$, first expand or contract the original signal $x(t)$ according to the value of α, then shift this intermediate result by β/α.



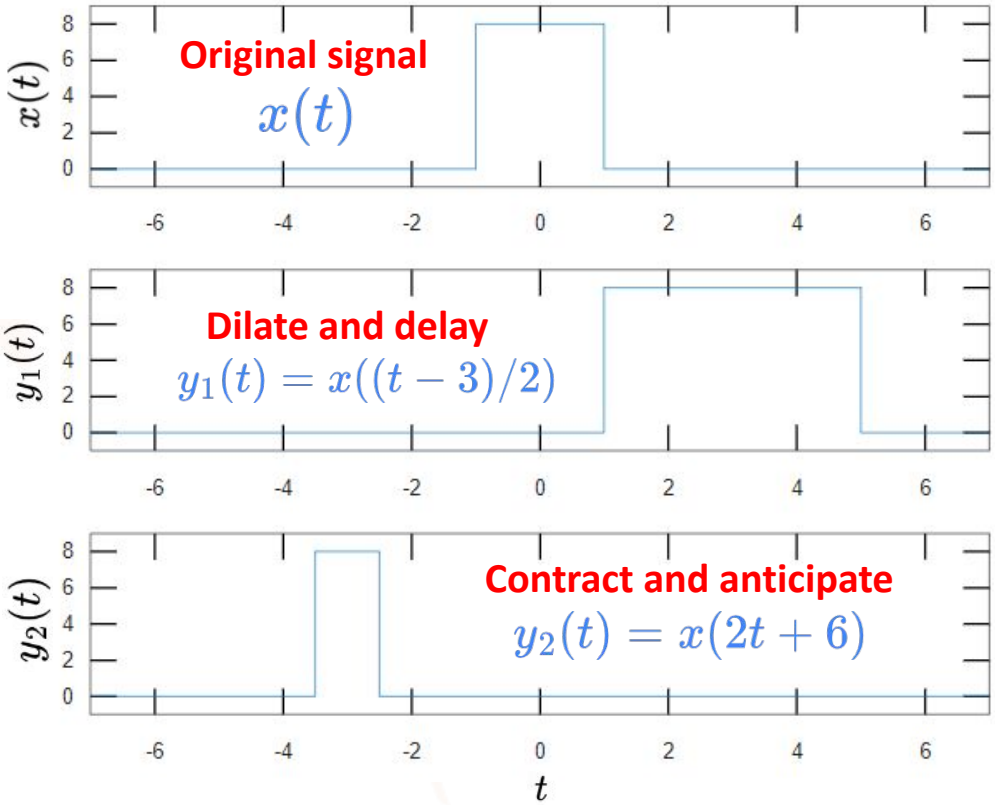


Two examples: expand+delay and contract+advance

Original signal $x(t)$: support from t = -1 to 1, with amplitude 8

$y_1(t) = x((t-3)/2)$: dilated the support by factor of 2, and shift center to t = 3. Now the support is from t = 1 to 5 (amplitude is 8, does not change)

$y_2(t) = x(2t+6) = x((t+3)/0.5)$: contract the support by factor of 2 (or "expand" by 0.5), and shift center to t = -3. Now the support is from t = -3.5 to -2.5



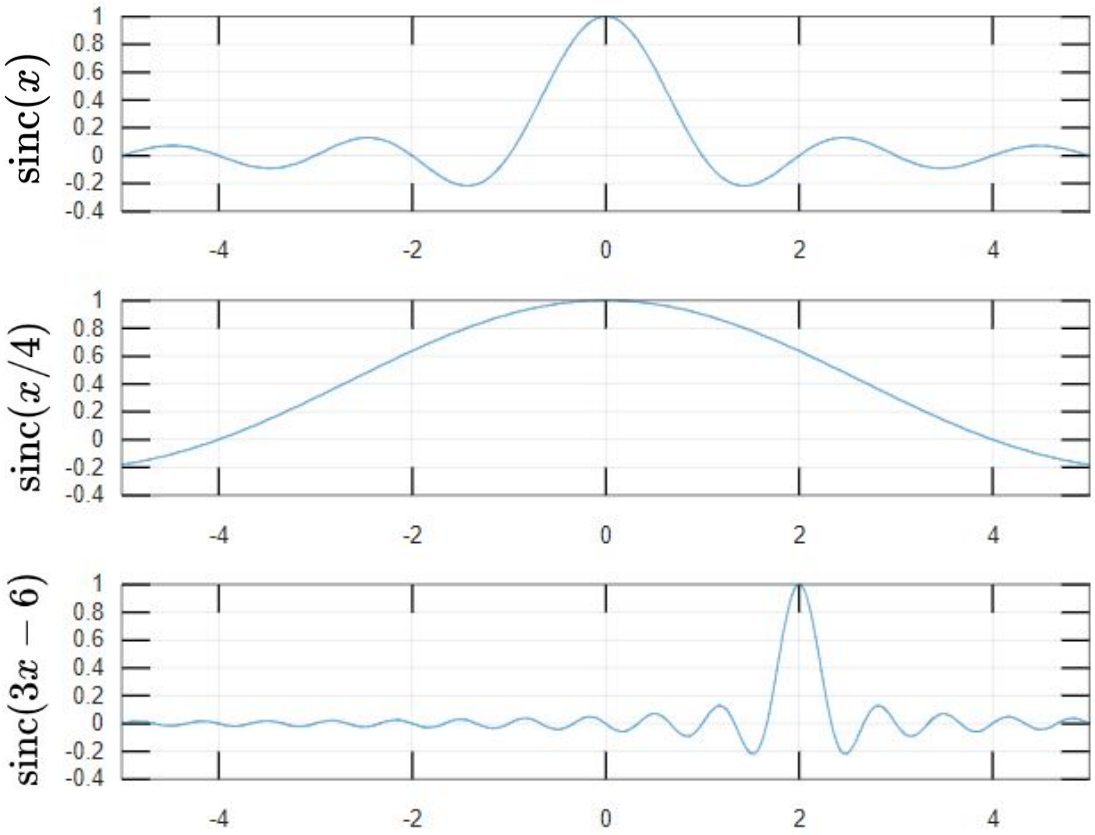
Example: sinc function

$$y_1(x) = \mathrm{sinc}(x/4)$$

First zero-crossing is now x = 4, while it is x = 1 for $\mathrm{sinc}(x)$, using the normalized version of sinc

$$y_2(x) = \mathrm{sinc}(3x - 6)$$
$$= \mathrm{sinc}\left(\frac{x-2}{1/3}\right)$$

First zero-crossing at x= 7/3 = 2.33

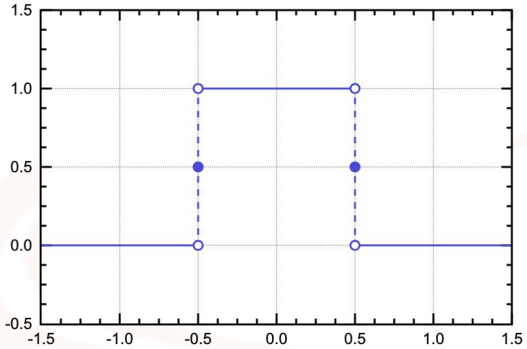


$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Now, can you understand the previous example ?

Example: rect function

Another important function is the rect or $\Pi(t)$, which corresponds to a normalized pulse with unitary support from $t = -0.5$ to 0.5 and amplitude equals to 1



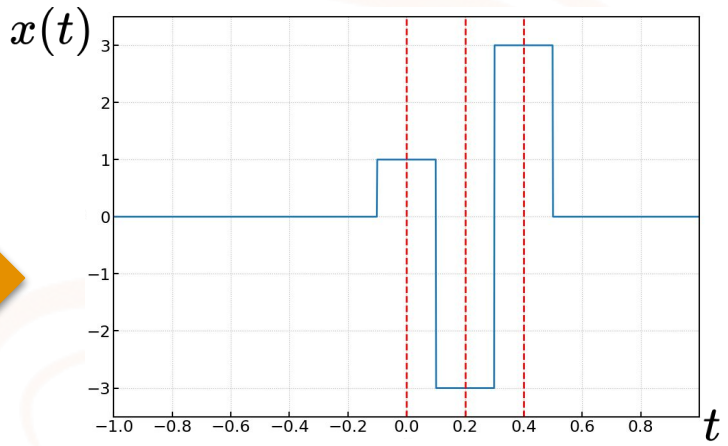
https://en.wikipedia.org/wiki/Rectangular_function

A rect can be described using an equation similar to that of the sinc function:

$$x(t) = A\text{rect}\left(\frac{t - t_c}{T}\right)$$

One good example of manipulating the rect signal is:

$$x(t) = \text{rect}\left(\frac{t}{0.2}\right) - 3\text{rect}\left(\frac{t - 0.2}{0.2}\right) + 3\text{rect}\left(\frac{t - 0.4}{0.2}\right)$$



When in doubt: use the TABLE

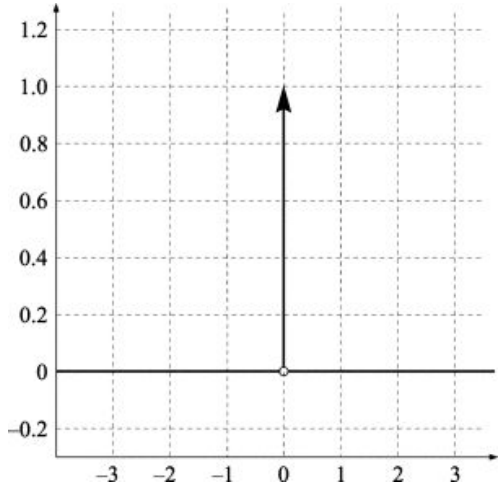
Using:

$$y_1[n] = x[n-2]$$
$$y_2[n] = x[-n+3]$$

n	$x[n]$	$n-2$	$y_1[n]$	$-n+3$	$y_2[n]$
-3	0	-5	0	6	36
-2	0	-4	0	5	25
-1	0	-3	0	4	16
0	0	-2	0	3	9
1	0	-1	0	2	0
2	0	0	0	1	0
3	9	1	0	0	0
4	16	2	0	-1	0
5	25	3	9	-2	0
6	36	4	16	-3	0
7	0	5	25	-4	0
8	0	6	36	-5	0
9	0	7	0	-6	0

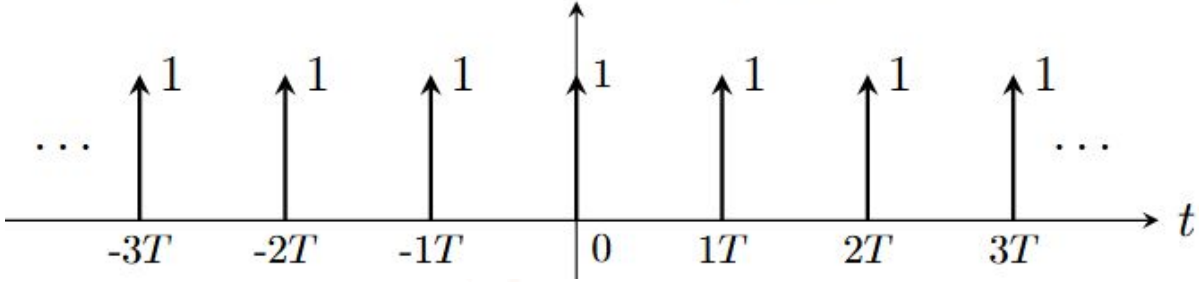
Another example of manipulating the independent variable: train of impulses

Continuous-time impulse **(Dirac)** function $\delta(t)$:
Amplitude is infinite at origin and zero elsewhere.
The impulse area is 1.



Can you understand the equation?

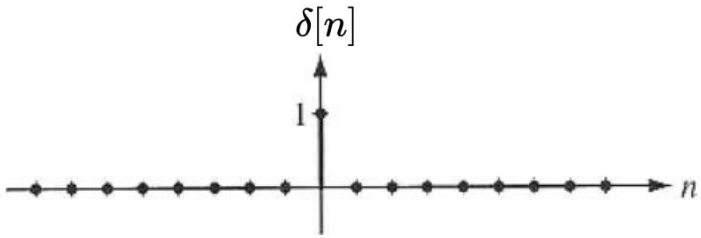
Continuous-time impulse train $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$

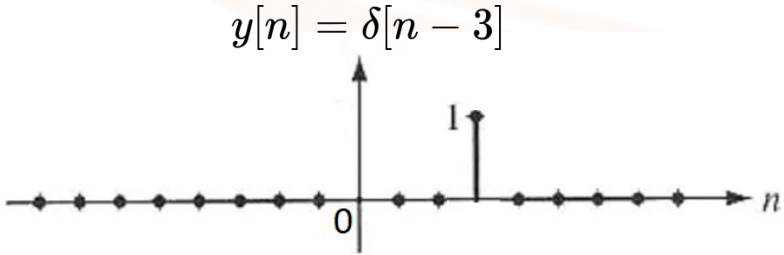


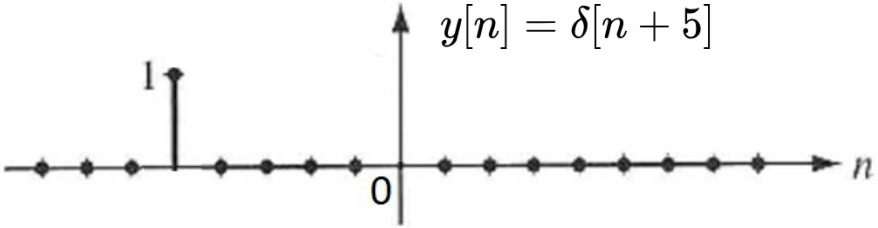
Discrete-time impulse function δ[*n*] (Kronecker's delta)

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

The amplitude is 1 at *n* = 0. There is no math issue (no "infinite amplitude") as in the continuous-time $\delta(t)$ case





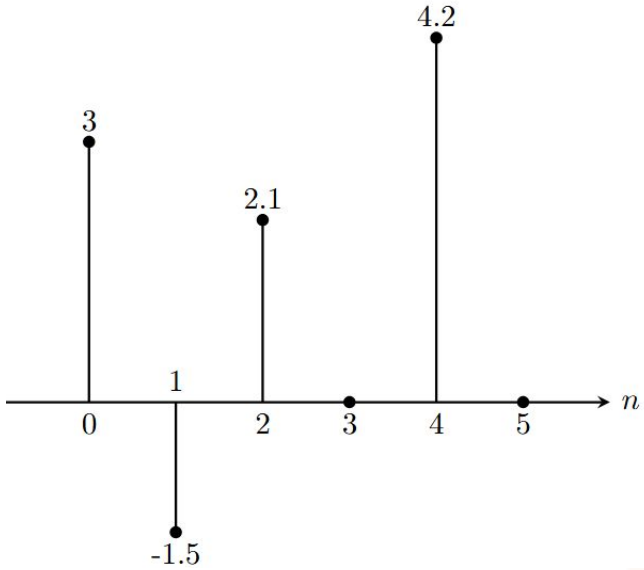


Time-shifted impulses are very useful

A great property of δ[*n*]

Any discrete-time signal can be conveniently represented by a sum of scaled and shifted impulses $\delta[n]$

Signal with 4 non-zero amplitudes can be represented with 4 impulses



This example: $x[n] = 3.0\delta[n] - 1.3\delta[n-1] + 2.1\delta[n-2] + 4.2\delta[n-4]$

In general: $$x[n] = \sum_{n_0=-\infty}^{\infty} x[n_0]\delta[n-n_0]$$

To represent 4 $\delta[n - n_0]$, the impulse $\delta[n - n_0]$ helps to locate the amplitude 4 at time n_0

How to mathematically represent a vector of samples stored in a computer?

Original vector with samples: $[3.0, -1.3, 2.1, 0, 4.2]$

Wrong way:

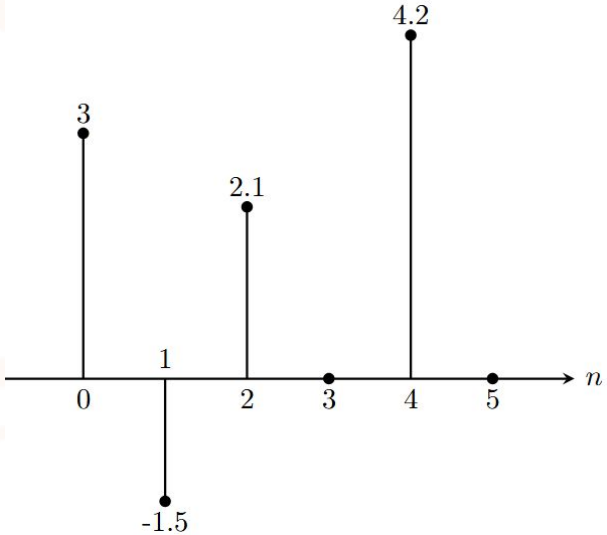
Values would be summed up!

$$x[n] = 3.0 - 1.3 + 2.1 + 0 + 4.2 \quad \Rightarrow \quad x[n] = 8$$

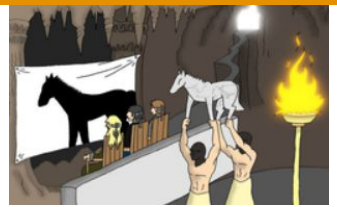
Correct way:

$$x[n] = 3.0\delta[n] - 1.3\delta[n-1] + 2.1\delta[n-2] + 4.2\delta[n-4]$$

The impulses $\delta[n - n_0]$ help to locate amplitudes in time.
We can build **any** discrete-time x[n] with impulses $\delta[n]$!
Can we build any continuous-time $x(t)$ with impulses $\delta(t)$?

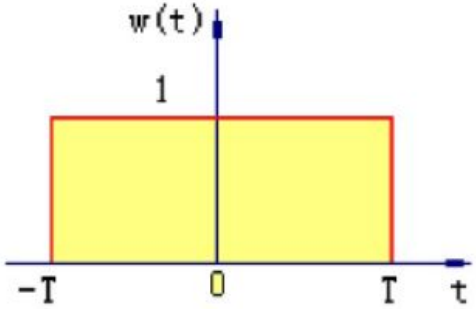


Windowing

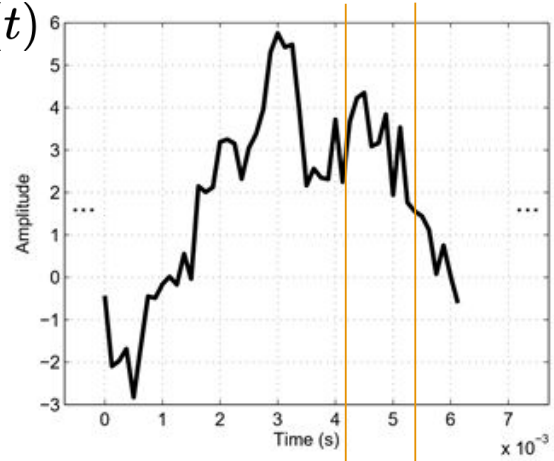


A window $w(t)$ is a signal of finite duration (support), which can be multiplied by the original signal $x(t)$, such that $y(t) = x(t)w(t)$ has zero amplitude outside the window support

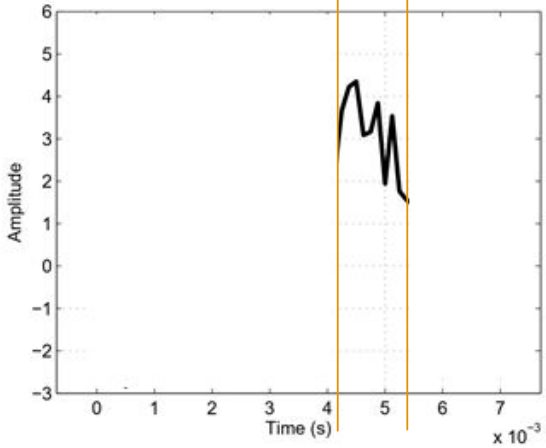
Rectangular window (there are many others)



$x(t)$



$y(t) = x(t)w(t)$



Sifting property of impulses

An impulse can be used as a window with extremely short duration: it “extracts” a single sample of the original signal

Sift □ analysis (it is not shift!)

Goal: understand expressions such as the ones below. Note that dealing with continuous-time impulses (with infinite amplitude, etc.) is more mathematically involved than in discrete-time

Generic expressions for any signal, using the impulses sifting property:

$$x[n] = \sum_{n_0=-\infty}^{\infty} x[n_0]\delta[n-n_0] \qquad x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(\tau - t)d\tau.$$

Signal sifting with discrete-time impulses

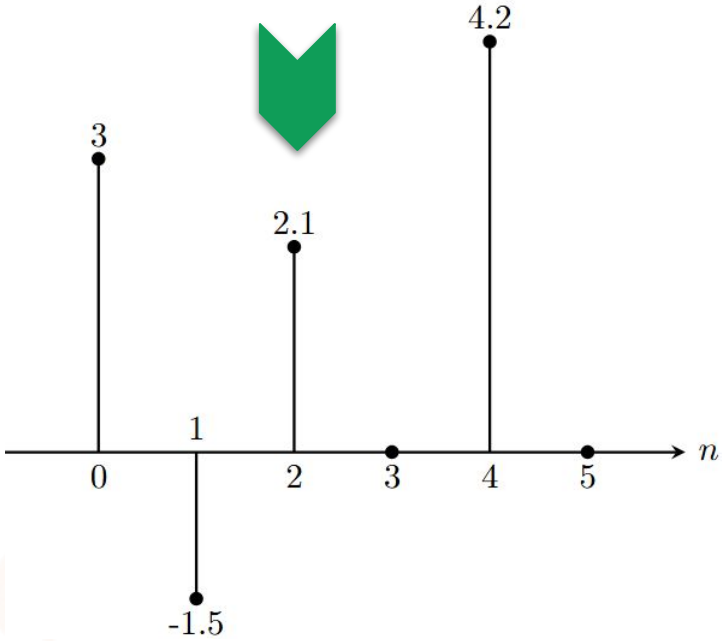
We can use the window: $w[n] = \delta[n-2]$ to observe $x[n]$ at n = 2

We get $y[n] = x[n]\delta[n-2] = 2.1\delta[n-2]$

How would you observe $x[n]$ at n = 4?

$x[n]\delta[n-n_0]$ is a "complete sequence" (informs amplitude for any n), with the amplitude $x[n_0]$ located at time by the impulse $\delta[n-n_0]$. It is not simply a "number"

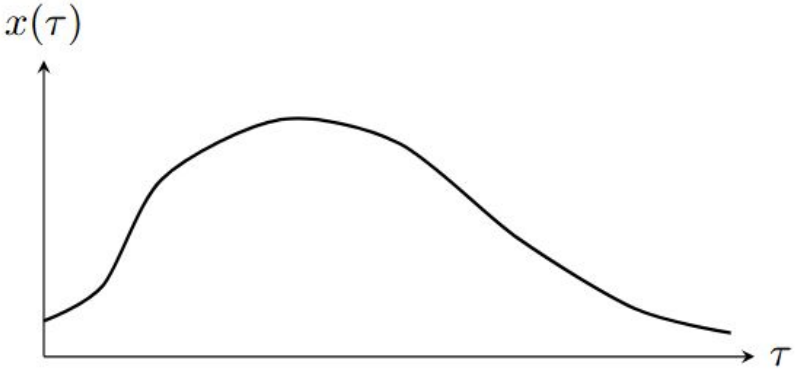
We want to "analyze" the signal at n = 2

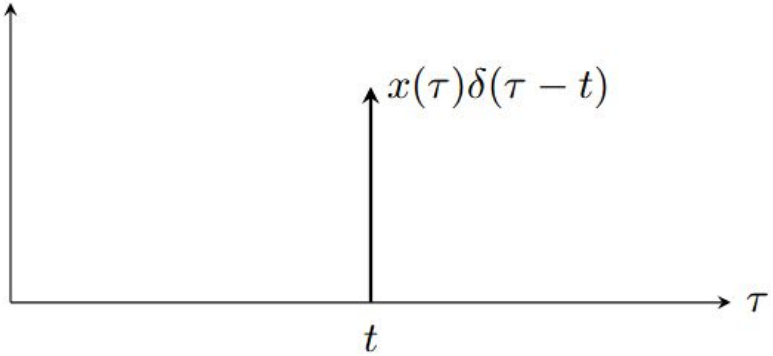


We can build any continuous-time x(t) with impulses δ(t)

Generic expression for any continuous-time signal using the impulses sifting property:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(\tau - t)d\tau$$





Who am I? A number, a sample, a "complete" signal, a murderer?

Beware: notation ambiguity!

A) $x(t)\delta(t-t_0)$

B) $x(t_0) = \int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt$

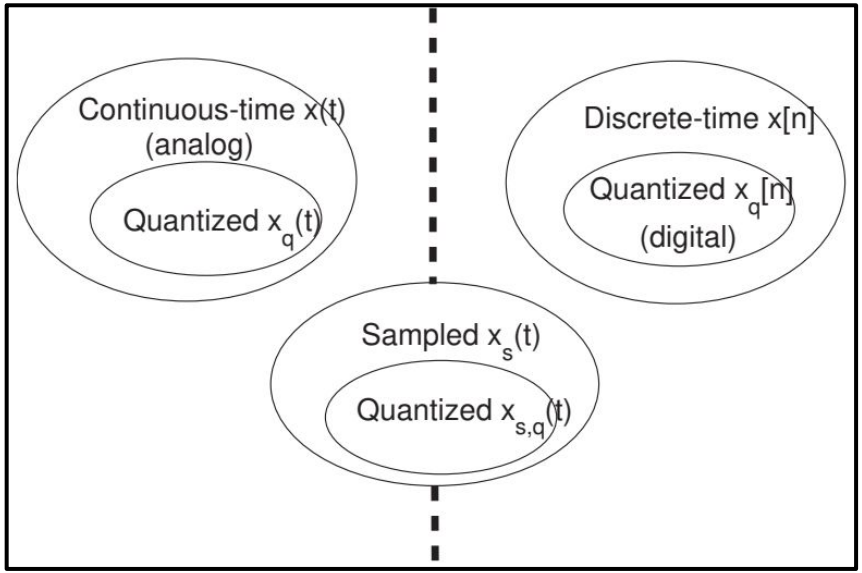
C) $x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(\tau-t)d\tau$

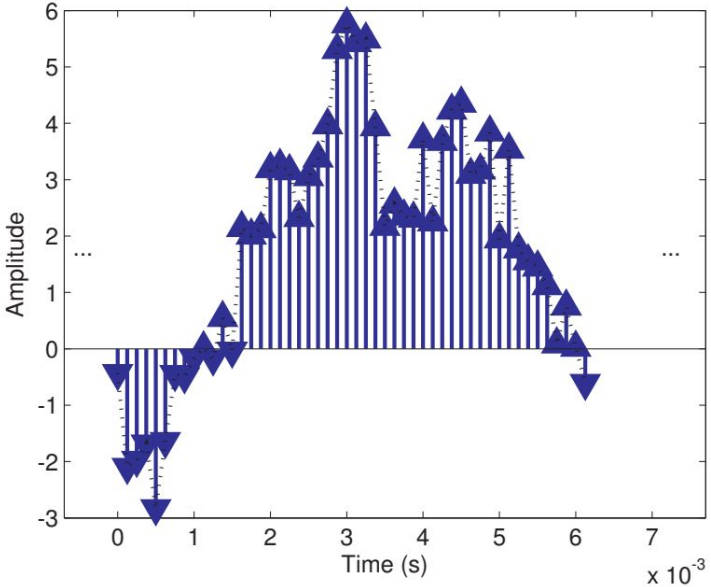
Assume t_0 is a real number and t the independent variable. How do you relate the questions with A), B) and C)?

1) A complete "signal", an expression to find the amplitude at any time instant
2) A single number, an amplitude, not explicitly located in time
3) Equivalent to $x(t_0)$
4) Equivalent to $x(t_0)\,\delta(t-t_0)$

Sampled signals: special continuous-time signals that can be seen as new category

A sampled signal $x_s(t)$ is assumed to be the result of periodic sampling

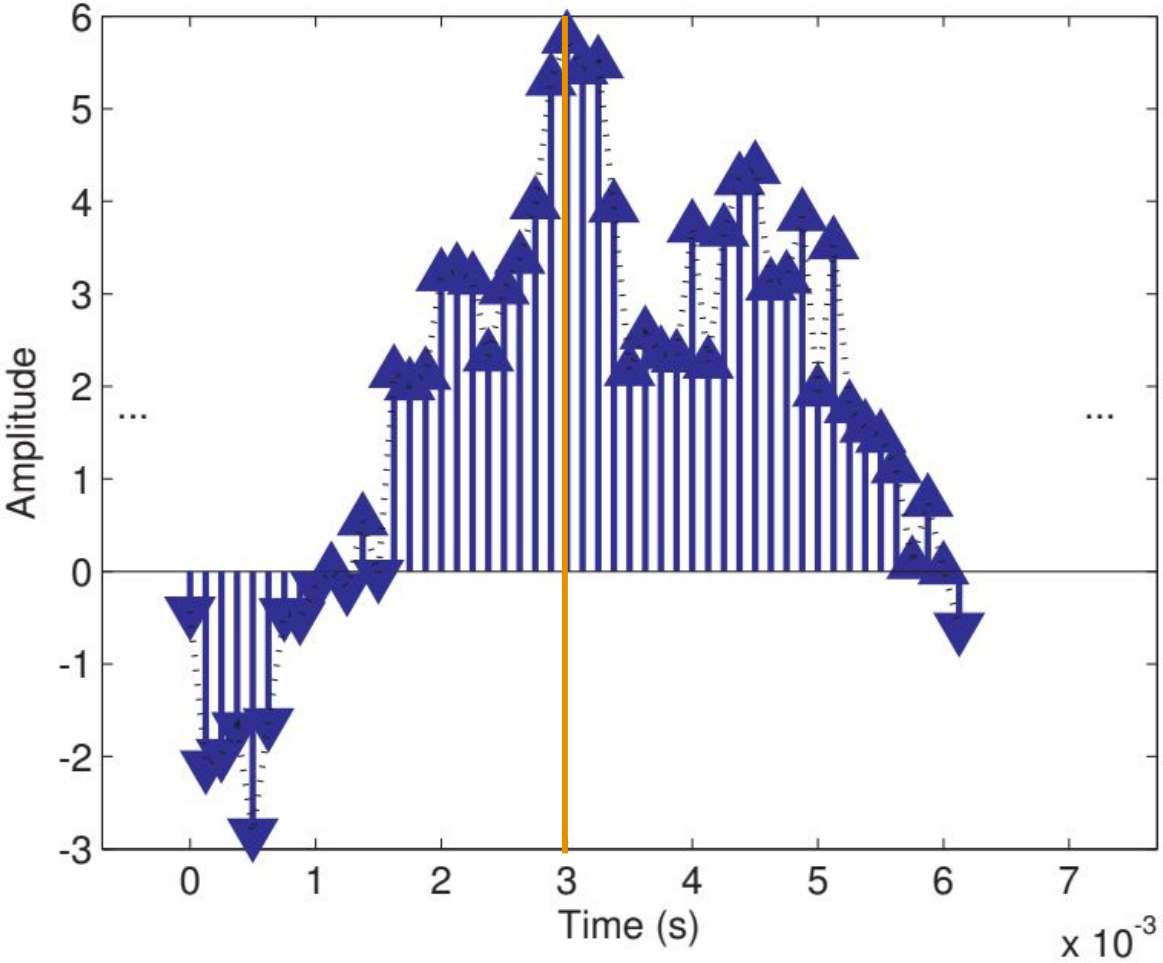




$T_s = 125\ \mu\text{s}$ (or, equivalently, that $F_s = 1/T_s = 8$ kHz)

Recall: amplitude of sampled signals is...

What is the value of:

$$x_s(t_0)$$

for t_0 = 3 ms?